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1 Cross-Dataset Embedding Model Comparison Report

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1.1 Executive Summary

1.1.1 Datasets Analyzed

- **education**: 1 PDF → 189 chunks, 5 queries, 21 qrels
- **finance**: 1 PDF → 50 chunks, 5 queries, 17 qrels
- **Total**: 2 datasets, 2 PDFs → 239 chunks, 10 queries

1.1.2 Overall Best Performers (by Average Rank)

1. **all-MiniLM-L6-v2** (Avg Rank: 1.00)
   - Best on: finance (MAP=0.84), education (MAP=0.80)
   - Consistency: CV=0.297
2. **all-MiniLM-L12-v2** (Avg Rank: 2.00)

- Best on: finance (MAP=0.75), education (MAP=0.51)
- Consistency: CV=0.379

3. **all-mpnet-base-v2** (Avg Rank: 3.50)
   - Best on: finance (MAP=0.54), education (MAP=0.43)
   - Consistency: CV=0.353
4. **paraphrase-multilingual-MiniLM-L12-v2** (Avg Rank: 3.50)
   - Best on: finance (MAP=0.56), education (MAP=0.38)
   - Consistency: CV=0.389

1.1.3 Key Findings

- **Statistical Significance**: all-MiniLM-L6-v2 does not significantly differ from all-MiniLM-L12-v2 on MAP (p=0.299)
- **Dataset Difficulty**: education dataset is harder (mean MAP=0.53 vs finance=0.67)
- **Performance Variability**: all-MiniLM-L6-v2 shows most consistent performance (CV=0.30)

1.2 Methodology

1.2.1 Evaluation Setup

- **Retrieval Method**: ChromaDB vector search with cosine similarity
- **k Value**: 10 (for all metrics)
- **Chunking**: 1000 tokens with 200-token overlap
- **Metrics Computed**: MAP, MRR, NDCG, PRECISION, RECALL

1.2.2 Statistical Analysis Methods

- **Paired t-tests**: Comparing models across datasets (alpha=0.05)
- **Effect Size**: Cohen's d for practical significance
- **Consistency Metrics**: Coefficient of variation, worst-case drop
- **Friedman Test**: chi-squared=5.40, p=0.145

1.3 Dataset Characteristics

Dataset	PDFs	Chunks	Queries	Qrels
education	1	189	5	21
finance	1	50	5	17
Total	2	239	10	38

Table: Dataset Statistics Summary

Column Explanations: - **Dataset**: Dataset identifier used in the benchmark - **PDFs**: Number of source PDF documents - **Chunks**: Total text chunks created from PDFs - **Queries**: Number of evaluation queries generated - **Qrels**: Number of relevance judgments (query-document labels)

How to interpret: - **More chunks** generally indicate larger corpora and higher retrieval complexity - **More queries and qrels** increase evaluation coverage and statistical power

Dataset	Avg Chunk Length	Avg Query Length	Avg Relevant Docs	Coverage Ratio
education	158.3	12.8	4.2	11.11%
finance	146.2	11.8	3.4	34.00%
Average	152.3	12.3	3.8	22.56%

Table: Aggregated Dataset Characteristics

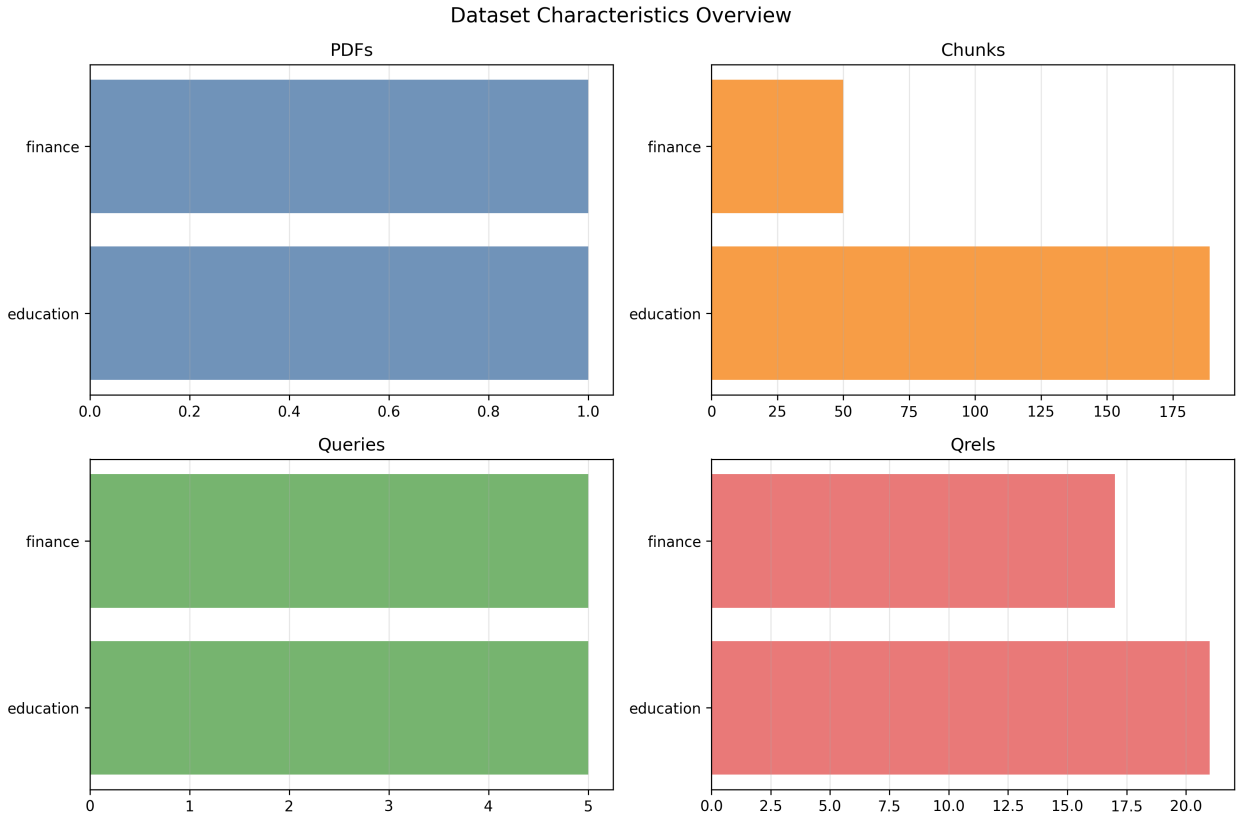


Figure 1: Dataset Characteristics Overview

Figure: Dataset Characteristics Overview

This figure compares dataset size indicators (PDFs, chunks, queries, qrels) across all datasets.

1.4 Cross-Dataset Model Comparison

1.4.1 Overall Performance Rankings (by MAP)

Model	Avg Rank	Mean MAP	Std	CV	Datasets Won
all-MiniLM-L6-v2	1	0.8209	0.0297	0.297	2
all-MiniLM-L12-v2	2	0.6287	0.1678	0.379	0
all-mpnet-base-v2	3.5	0.4872	0.077	0.353	0
paraphrase-multilingual-MiniLM-L12-v2	3.5	0.4701	0.1244	0.389	0

Table: Cross-Dataset Model Performance Summary

Column Explanations: - **Avg Rank**: Average ranking position across datasets (1.0 = best overall) - **Mean MAP**: Average MAP score across all datasets - **Std**: Standard deviation of performance across datasets - **CV**: Coefficient of variation (Std/Mean) - measures relative consistency - **Datasets Won**: Number of datasets where this model ranked #1

How to use this table: - **Sort by Avg Rank**: Identifies the most reliable models overall - **Check CV values**: Lower CV indicates more consistent performance - **Compare Mean vs Std**: High mean with low std = reliable excellence - **Datasets Won**: Shows how often a model is the best choice

1.4.2 Model Performance by Dataset (MAP)

Model	education	finance
all-MiniLM-L12-v2	0.5101	0.7474
all-MiniLM-L6-v2	0.7999	0.8419
all-mpnet-base-v2	0.4328	0.5417
paraphrase-multilingual-MiniLM-L12-v2	0.3821	0.558

Figure: Model Performance by Dataset

This heatmap shows MAP scores for each model across all datasets. Darker colors indicate stronger performance.

1.4.3 Embedding Speed Comparison (Docs/sec)

Model	education	finance
all-MiniLM-L12-v2	47.8	47.9
all-MiniLM-L6-v2	45.1	45.9
all-mpnet-base-v2	4.1	4.8
paraphrase-multilingual-MiniLM-L12-v2	47.3	47.9

Figure: Embedding Throughput by Dataset

This heatmap compares embedding throughput in documents per second across datasets.

Figure: Cross-Dataset Performance Comparison

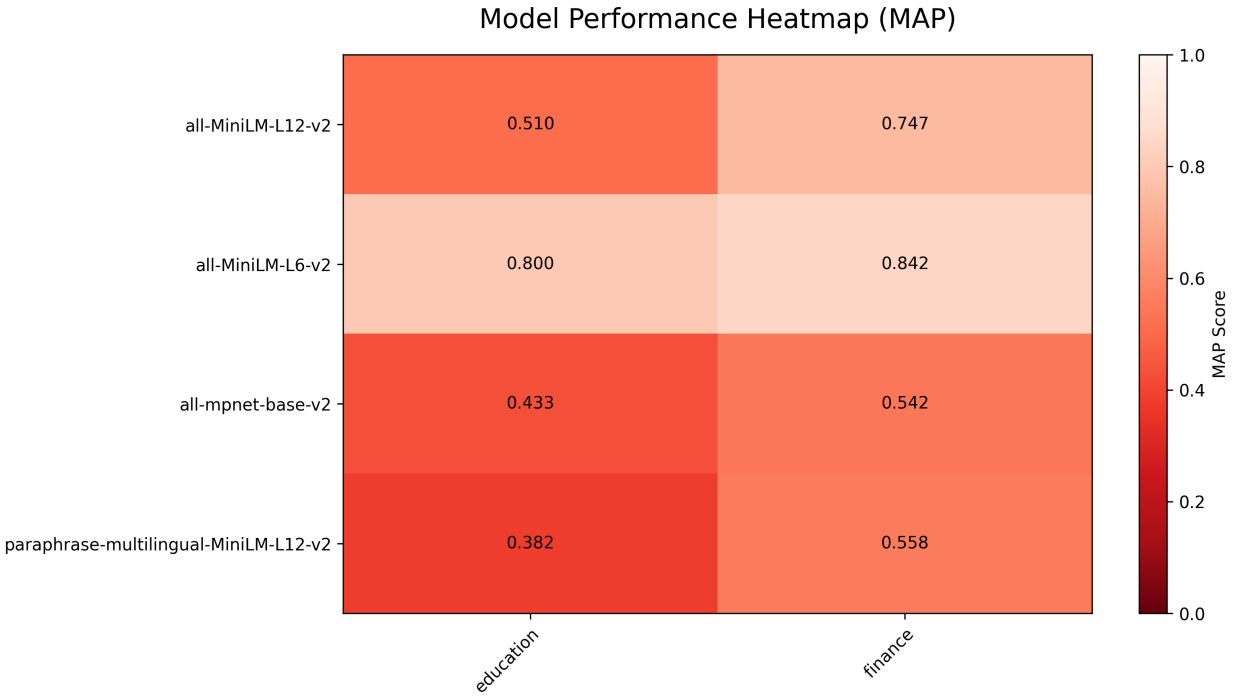


Figure 2: Model Performance Heatmap

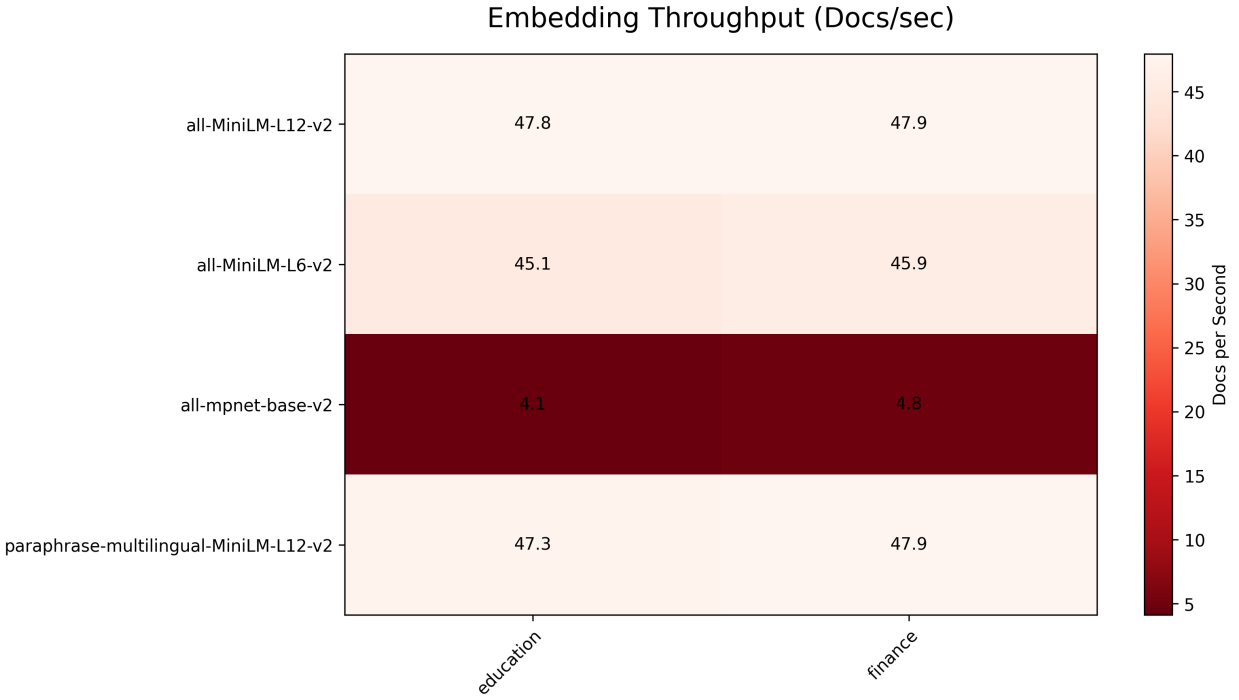


Figure 3: Embedding Speed Heatmap

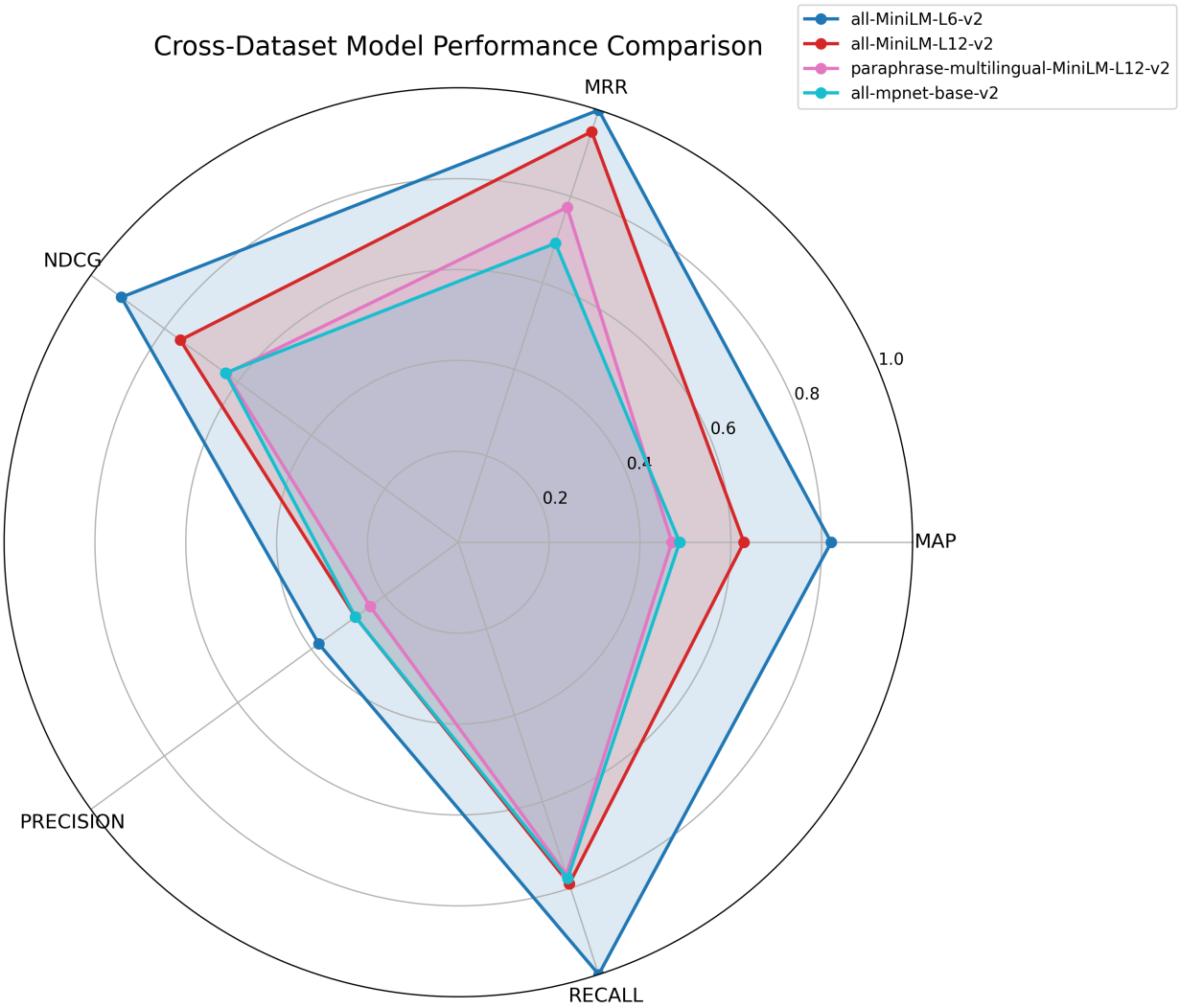


Figure 4: Cross-Dataset Radar Chart

This radar chart shows how each embedding model performs across all evaluated metrics (MAP, NDCG, etc.) and datasets simultaneously. Each axis represents a different metric, with values normalized to a common scale.

How to interpret: - **Larger area**: Better overall performance across metrics - **Balanced shape**: Consistent performance across different evaluation aspects - **Spiky shape**: Strong in some metrics but weak in others - **Position relative to center**: Distance from center indicates performance level

Why it matters: Identifies models with well-rounded capabilities vs. specialized models, helping you choose based on your specific requirements.

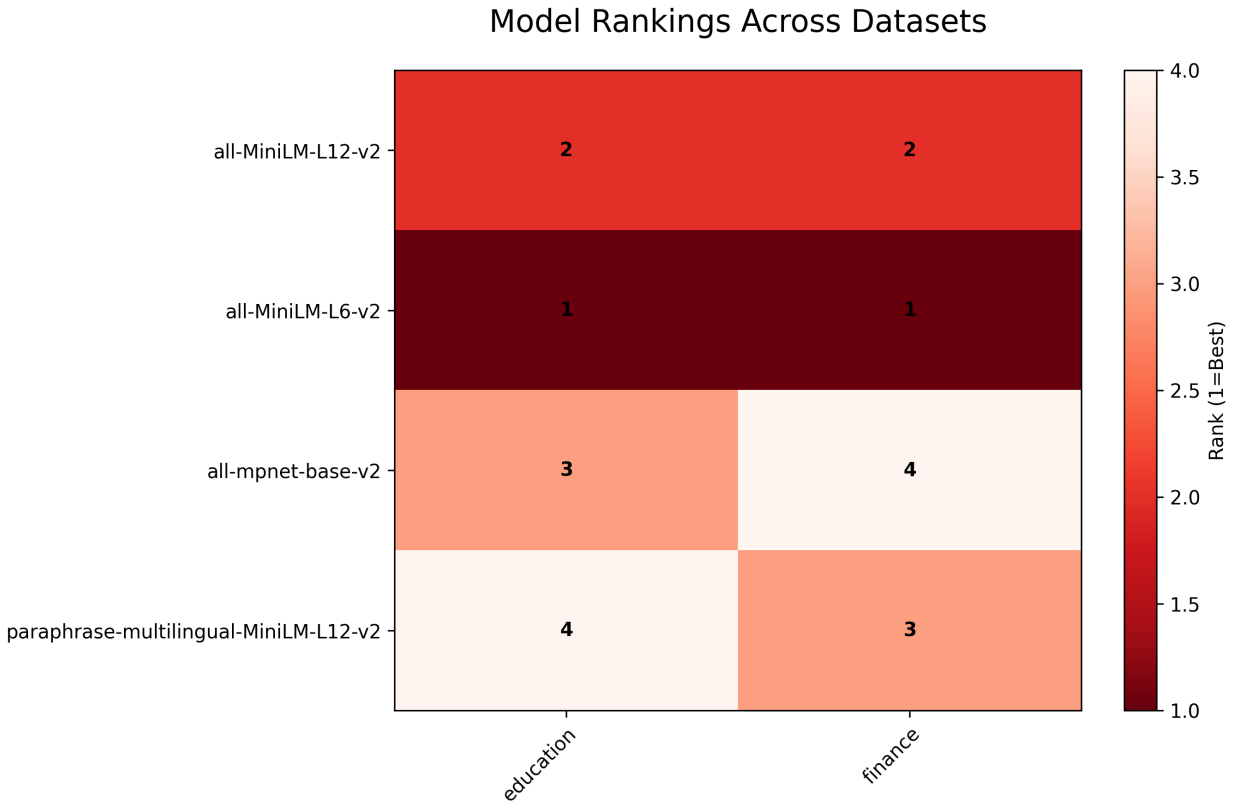


Figure 5: Model Ranking Heatmap

Figure: Model Rankings Across Individual Datasets

This heatmap displays the ranking position (1=best, higher numbers=worse) of each model within each dataset. Darker colors indicate better performance (lower ranks).

How to interpret: - **Dark horizontal stripes**: Model performs consistently well across datasets - **Varied colors per model**: Performance varies significantly by dataset - **Light colors**: Model struggles on those specific datasets - **Column patterns**: Reveals which datasets are easier/harder for all models

Why it matters: Shows model reliability across different domains and identifies specialized vs. general-purpose models.

1.4.4 Statistical Significance Analysis

This section determines whether observed performance differences between models are statistically significant (real differences) rather than random chance.

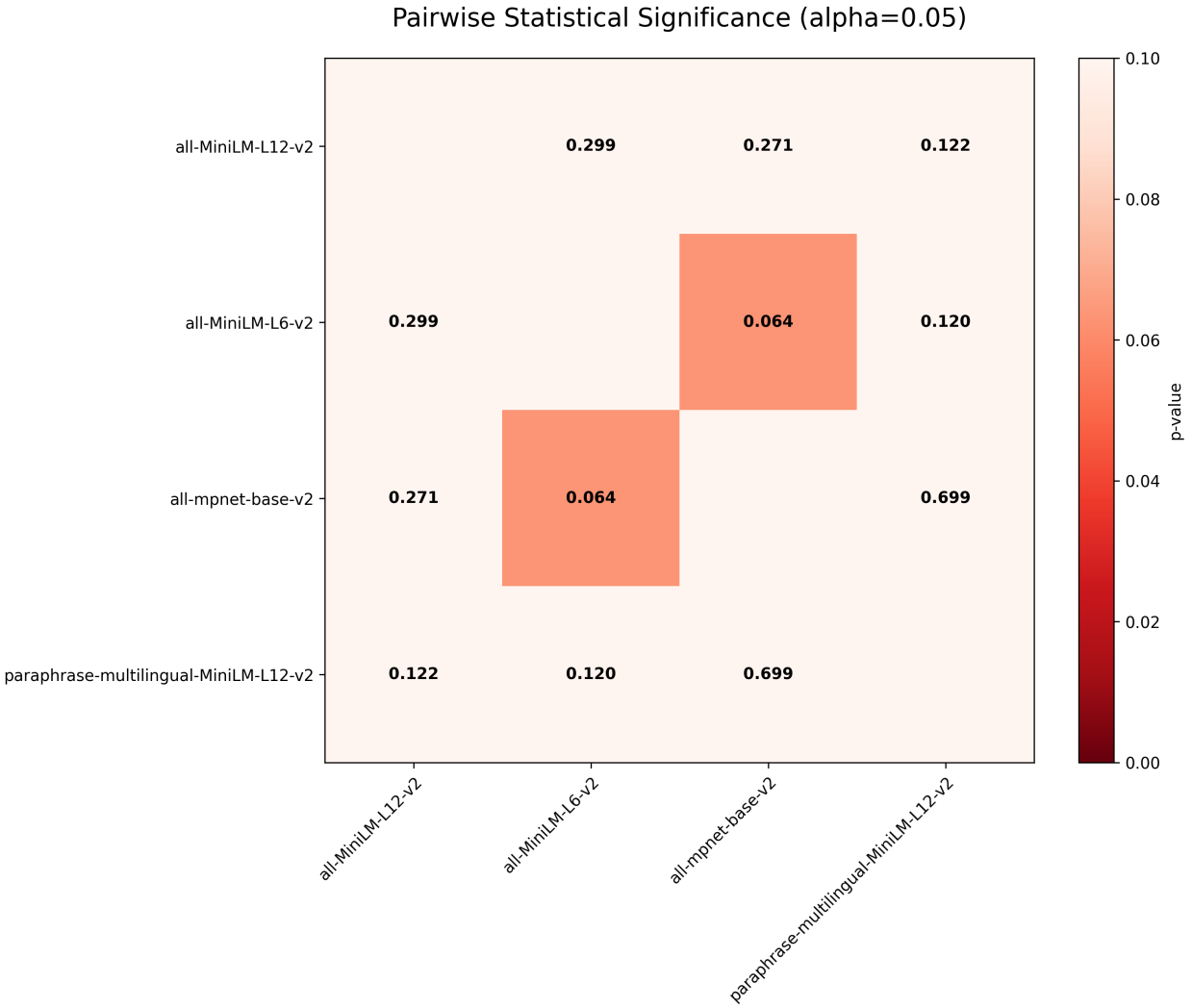


Figure 6: Pairwise Significance Heatmap

Figure: Statistical Significance of Performance Differences

This heatmap shows p-values from paired t-tests comparing each model pair across all datasets. Darker colors indicate more significant differences.

Statistical terms explained: - **p-value**: Probability that observed differences occurred by chance - **alpha**: Significance threshold (0.05 = 5% chance) - **$p < 0.05$**: Differences are statistically significant - **$p >= 0.05$**: Differences may be due to random variation

How to interpret: - **Dark cells**: Strong evidence that models truly perform differently - **Light cells**: Models perform similarly (differences likely random) - **Diagonal**: Always white (comparing model to itself)

Why it matters: Ensures you don't choose a model based on noise. Only act on differences that are statistically significant.

Interpretation Guide: * p<0.05, ** p<0.01, *** p<0.001, ns = not significant

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1.5 Dataset-Specific Analysis

1.5.1 Education Dataset

- **PDFs**: 1 original document
- **Chunks**: 189 text chunks (avg 189.0 per PDF)
- **Queries**: 5 queries, 21 relevance judgments
- **Difficulty**: Medium (mean MAP=0.53 across models)
- **Best Model**: all-MiniLM-L6-v2 (MAP=0.7999)

1.5.1.1 Corpus Characteristics

Metric	Value
Source PDF Documents	1
Total Document Chunks	189
Average Chunk Length (words)	158.3
Median Chunk Length (words)	144
Min Chunk Length (words)	67
Max Chunk Length (words)	381
Standard Deviation	49.5

1.5.1.2 Query Characteristics

Metric	Value
Total Queries	5
Average Query Length (words)	12.8
Median Query Length (words)	12
Min Query Length (words)	7
Max Query Length (words)	19
Standard Deviation	3.9

1.5.1.3 Relevance Distribution

Metric	Value
Total Relevance Judgments	21
Average Relevant Docs per Query	4.2
Median Relevant Docs per Query	4.0
Min Relevant Docs	2
Max Relevant Docs	9

Metric	Value
Coverage Ratio	11.11%

1.5.1.4 Representative Query Examples

Query ID	Query Text	# Relevant Docs
q1_for_DeepSeek-V3_chunk_85	What are the DeepSeek-V3 training hyperparameters and learning rate schedule	2
q2_for_DeepSeek-V1_chunk_109	What are the methods used to enhance model responses through reflection and verification, and how is non-reasoning data handled?	2
q3_for_DeepSeek-V3_chunk_21	What are the key components and training methodologies of the DeepSeek-V3 model?	4
q4_for_DeepSeek-V3_chunk_84	What is the Multi-Token Prediction (MTP) objective and its implementation in DeepSeek-V3?	4
q5_for_DeepSeek-V3_chunk_52	What is a mixed precision framework for FP8 data format in deep learning training?	9

1.5.1.5 Embedding Performance Metrics

Model	MAP@10	MRR@10	NDCG@10	PRECISION@10	RECALL@10
all-MiniLM-L12-v2	0.5101	0.9	0.6507	0.28	0.6556
all-MiniLM-L6-v2	0.7999	1	0.9092	0.42	1
all-mpnet-base-v2	0.4328	0.5333	0.5936	0.32	0.8056
paraphrase-multilingual-MiniLM-L12-v2	0.3821	0.7167	0.558	0.24	0.6889

1.5.1.6 Embedding Speed Comparison

Model	Embedding Time (s)	Docs Processed	Docs/sec	Avg Time per Doc (ms)
all-MiniLM-L12-v2	3.96	189	47.78	20.93
all-MiniLM-L6-v2	4.19	189	45.08	22.18
all-mpnet-base-v2	45.92	189	4.12	242.96
paraphrase-multilingual-MiniLM-L12-v2	4	189	47.31	21.14

1.5.1.7 Dataset Summary Notes

- Best model: all-MiniLM-L6-v2 (MAP=0.7999)
- Mean MAP across models: 0.5312
- Timing data available for 4 model(s)

1.5.2 Finance Dataset

- **PDFs**: 1 original document
- **Chunks**: 50 text chunks (avg 50.0 per PDF)
- **Queries**: 5 queries, 17 relevance judgments
- **Difficulty**: Medium (mean MAP=0.67 across models)
- **Best Model**: all-MiniLM-L6-v2 (MAP=0.8419)

1.5.2.1 Corpus Characteristics

Metric	Value
Source PDF Documents	1
Total Document Chunks	50
Average Chunk Length (words)	146.2
Median Chunk Length (words)	145
Min Chunk Length (words)	67
Max Chunk Length (words)	193
Standard Deviation	19.8

1.5.2.2 Query Characteristics

Metric	Value
Total Queries	5
Average Query Length (words)	11.8
Median Query Length (words)	12
Min Query Length (words)	6
Max Query Length (words)	18
Standard Deviation	4.7

1.5.2.3 Relevance Distribution

Metric	Value
Total Relevance Judgments	17
Average Relevant Docs per Query	3.4
Median Relevant Docs per Query	2.0
Min Relevant Docs	1
Max Relevant Docs	8
Coverage Ratio	34.00%

1.5.2.4 Representative Query Examples

Query ID	Query Text	# Relevant Docs
q1_for_Attention_chunk0	What are some key papers in sequence learning and deep learning, including work on LSTMs and residual networks?	4
q2_for_Attention_chunk0	Transformer model architecture for sequence transduction	8
q3_for_Attention_chunk25	What are the training data, hardware, and schedule used for language models?	1
q4_for_Attention_chunk28	What are the dropout and label smoothing techniques used in the Transformer model for machine translation?	2
q5_for_Attention_chunk49	What is the ideal application of law?	2

1.5.2.5 Embedding Performance Metrics

Model	MAP@10	MRR@10	NDCG@10	PRECISION@10	RECALL@10
all-MiniLM-L12-v2	0.7474	1	0.8627	0.28	0.925
all-MiniLM-L6-v2	0.8419	1	0.9252	0.34	1
all-mpnet-base-v2	0.5417	0.85	0.6742	0.24	0.75
paraphrase-multilingual-MiniLM-L12-v2	0.558	0.8333	0.705	0.24	0.85

1.5.2.6 Embedding Speed Comparison

Model	Embedding Time (s)	Docs Processed	Docs/sec	Avg Time per Doc (ms)
all-MiniLM-L12-v2	1.04	50	47.94	20.86
all-MiniLM-L6-v2	1.09	50	45.92	21.78
all-mpnet-base-v2	10.31	50	4.85	206.23
paraphrase-multilingual-MiniLM-L12-v2	1.04	50	47.93	20.86

1.5.2.7 Dataset Summary Notes

- Best model: all-MiniLM-L6-v2 (MAP=0.8419)
- Mean MAP across models: 0.6723
- Timing data available for 4 model(s)

Figure: Dataset Difficulty Assessment

This chart shows the average MAP performance across all models for each dataset. Higher bars indicate easier datasets (better overall performance), lower bars indicate harder datasets.

How to interpret: - **Tall bars**: Datasets where models perform well overall (easier retrieval tasks) - **Short bars**: Datasets where all models struggle (harder retrieval tasks) - **Consistent heights**: Similar difficulty across datasets - **Variable heights**: Some datasets are much harder than others

Why it matters: Identifies which datasets are good benchmarks and helps explain why some models might appear better (they excel on easier datasets).

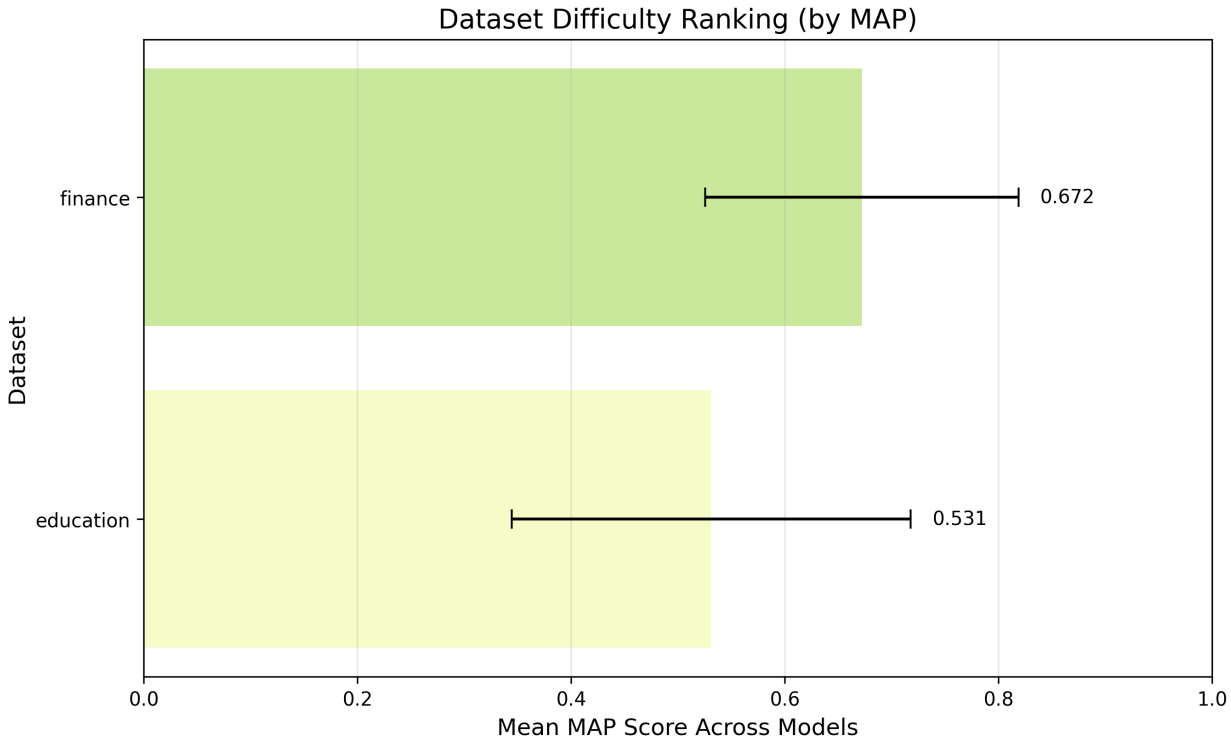


Figure 7: Dataset Difficulty Analysis

1.5.3 Dataset Summary

Dataset	Best Model	Top MAP	Models Tested	Difficulty
education	all-MiniLM-L6-v2	0.7999	4	0.47
finance	all-MiniLM-L6-v2	0.8419	4	0.33

Table: Dataset Performance Summary

Column Explanations: - **Dataset**: Name of the dataset evaluated - **Best Model**: The embedding model that achieved the highest score - **Top MAP**: The highest MAP score achieved on this dataset - **Models Tested**: Number of embedding models evaluated - **Difficulty**: Relative difficulty score (higher = more challenging)

How to use this table: - **Identify dataset characteristics**: Easy datasets have high top scores, hard ones have low scores - **Model selection guidance**: See which models excel on specific dataset types - **Benchmark planning**: Use difficulty scores to prioritize dataset development

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1.6 Model Consistency Across Datasets

This section measures how reliably each model performs across different datasets. A consistent model delivers similar performance regardless of the domain.

Figure: Model Performance Consistency

This bar chart shows two key consistency metrics for each model: - **Coefficient of Variation (CV)**: Standard deviation divided by mean (lower = more consistent) - **Worst Case Drop**: Maximum performance decrease from best to worst dataset

How to interpret: - **Low CV values**: Model performs similarly across all datasets (reliable) - **High CV values**: Performance varies greatly by dataset (unreliable) - **Small worst-case drop**:

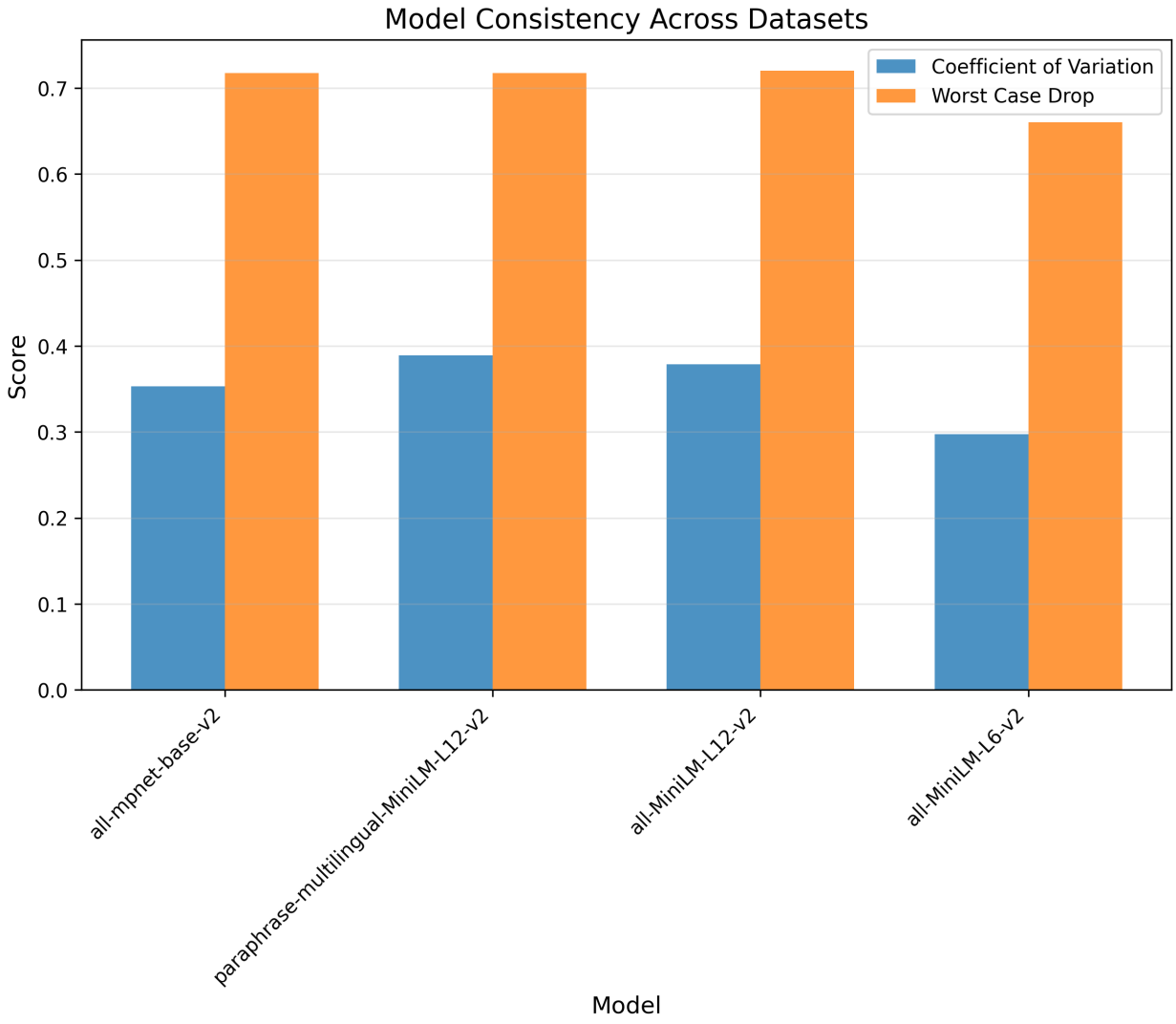


Figure 8: Consistency Comparison Chart

Worst Case Drop: Maximum performance decrease from best to worst dataset. Values > 0.3 indicate significant performance degradation on some datasets.

Score Range: Total spread between best and worst dataset performance. Smaller ranges indicate more reliable models.

1.7 Recommendations

1.7.1 For Production Deployment

1. **General Purpose**: Use all-MiniLM-L6-v2
   - Best overall performance (avg rank 1.00)
   - Strong performance (mean MAP=0.8209)

1.7.2 Limitations

- Evaluation limited to 2 datasets
- Single k value (10) - performance may vary at different k
- All datasets are document retrieval tasks
- Statistical power limited by number of datasets

1.8 Appendix

1.8.1 A. Raw Data References

- **education**: `datasets\education\reports\raw_results_2026-02-07_15-14-37.json`
- **finance**: `datasets\finance\reports\raw_results_2026-02-07_15-12-28.json`

1.8.2 B. Configuration

- Significance Level: alpha = 0.05
- Minimum Datasets Required: 2
- Statistical Tests Enabled: True
- Visualizations Enabled: True

Report generated by RagRetrievalBench Aggregate Reporting System